

Algebra 1
Unit 8
Lesson 1 Practice

Name _____
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1. The equations $y = -3(x + 2)(x - 3)$ and $y = -3\left(x - \frac{1}{2}\right)^2 + \frac{75}{4}$ both represent the same quadratic. Rewrite both equations in standard form, $y = ax^2 + bx + c$. Show your work.

$y = -3(x + 2)(x - 3)$	$y = -3\left(x - \frac{1}{2}\right)^2 + \frac{75}{4}$

2. The equations $y = -(x + 6)(x - 3)$ and $y = -(x - (-6))(x - 3)$ are equivalent. How can both x -intercepts be determined if the equation $y = -(x + 6)(x - 3)$ is given?

3. The equation $y = -2x^2 - 4x + 30$ can be used to determine the y -intercept. Show how each of the three forms can be used to find the y -intercept.

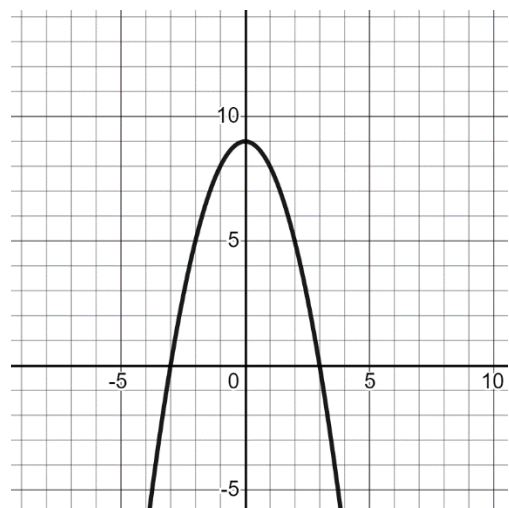
$y = -2x^2 - 4x + 30$	$y = -2(x + 5)(x - 3)$	$y = -2(x + 1)^2 + 32$

4. A table of values for $y = \frac{1}{4}(x - 2)^2 + 8$ is shown. What features can be found using the table of values?

- ☐ x -intercepts
- ☐ y -intercepts
- ☐ vertex
- ☐ axis of symmetry
- ☐ increasing/decreasing
- ☐ positive/negative

x	y
-3	14.25
-2	12
-1	10.25
0	9
1	8.25
2	8
3	8.25

5. The graph of a quadratic equation is shown. Use words, ordered pairs, or symbols to describe the location of each key feature.



Key Feature	Location of Key Feature
x -intercepts	
y -intercepts	
vertex	
axis of symmetry	
increasing	
decreasing	

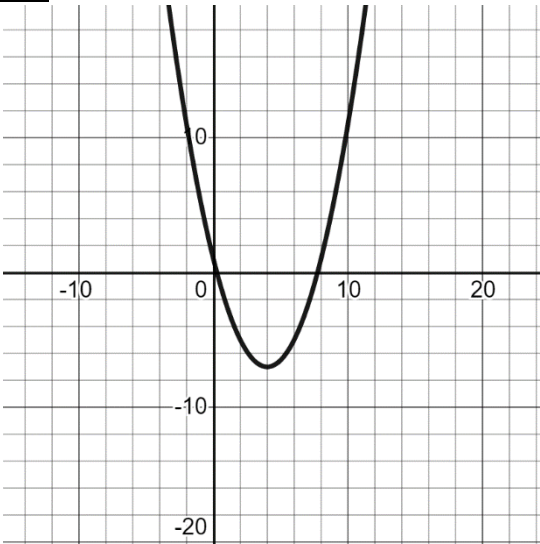
6. A table of values for $y = (x - 1)^2$ is shown. What features can be found using the table of values?

- ☐ x –intercepts
- ☐ y –intercepts
- ☐ vertex
- ☐ axis of symmetry
- ☐ increasing/decreasing
- ☐ positive/negative

x	y
-1	4
0	1
1	0
2	1
3	4

Use the graph of the quadratic equation for questions 7-10.

- 7. Describe where the parabola is increasing.
- 8. Describe where the parabola is decreasing.
- 9. Draw the axis of symmetry for the parabola on the graph.
- 10. Circle the vertex, x –intercepts, and y – intercepts of the parabola. Label each point as vertex, x –intercepts, and y – intercepts.



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Unit 8
Lesson 2 Practice

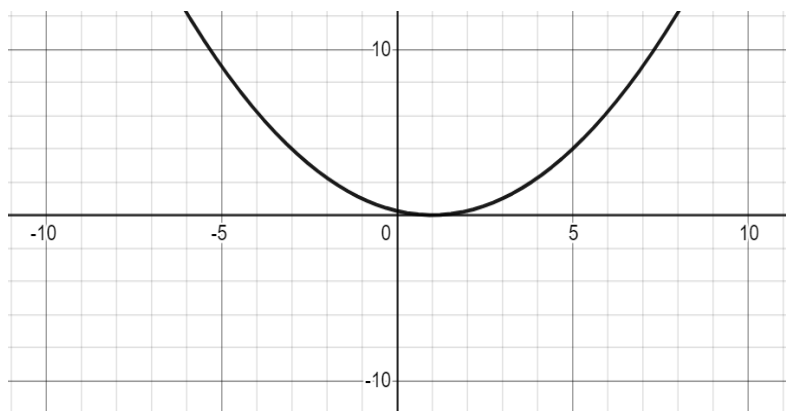
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1. As the x -values of the parabola get infinitely *larger*, the y -values of the parabola approach

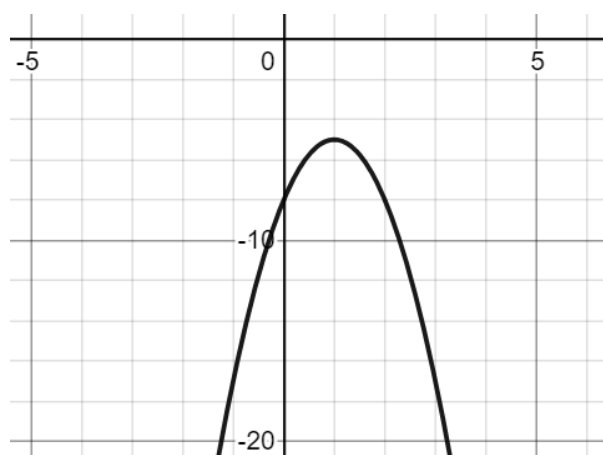
- ☐ ∞
☐ $-\infty$

2. As the x -values of the parabola get infinitely *smaller*, the y -values of the parabola approach

- ☐ ∞
☐ $-\infty$



Use the graph of the quadratic for questions 3 and 4.



3. Use words or symbols to describe the domain of the quadratic.

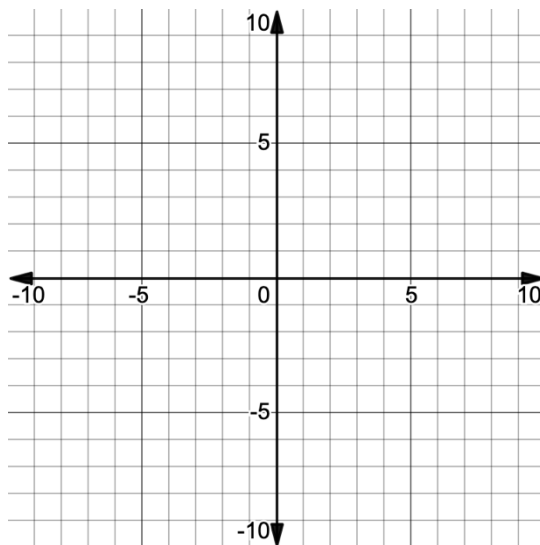
4. Use words or symbols to describe the range of the quadratic.

5. Determine the x -intercepts and y -intercepts of a quadratic using the equation $y = -\frac{4}{3}(x - 2)(x + 3)$.

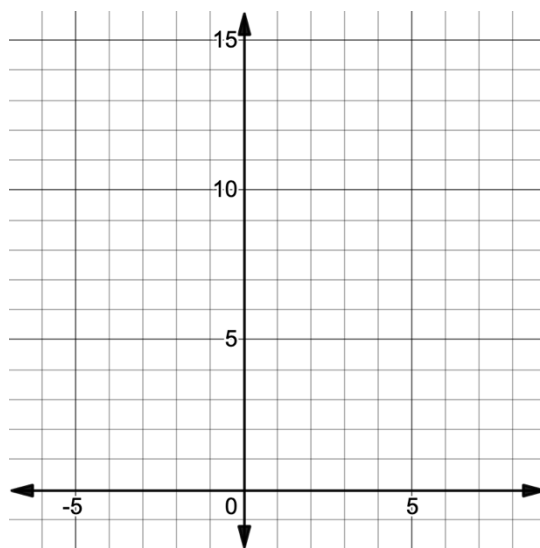
6. Use the table of values to determine the vertex of the parabola.

x	y
-2	5.333
-1.5	7
-1	8
-0.5	8.333
0	8
0.5	7
1	5.333

7. Use the key features you determined in question 5 and 6 to graph the parabola of the quadratic function.

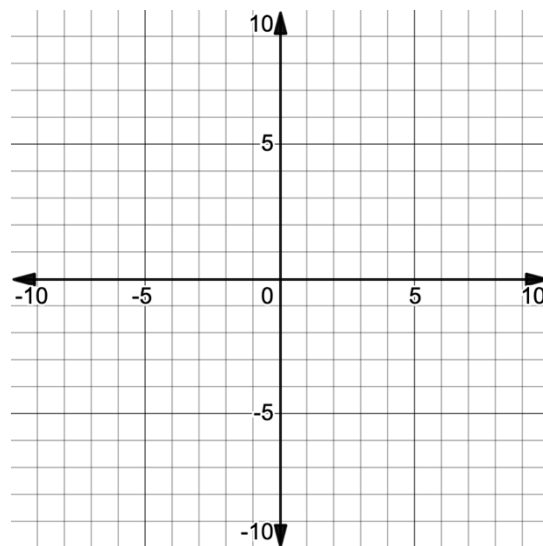


8. Use the equation $y = -(x - 7)(x + 1)$ to graph the quadratic.



9. Determine all of the key features possible using the equation of the quadratic $y = 2(x - 2)(x - 1)$.

10. Use the key features found in question 9 to graph the quadratic.



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Lesson 3 Practice

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Complete the table for each equation in 1-5.

Equation	Form	Key Feature(s) Revealed	Value(s) of Key Feature
1. $y = (x - 9)(x + 12)$	☐ Vertex form ☐ Factored form ☐ Standard form		
2. $y = (x - 5)^2 - 8$	☐ Vertex form ☐ Factored form ☐ Standard form		
3. $y = -\frac{1}{4}x^2 - 4x - 4$	☐ Vertex form ☐ Factored form ☐ Standard form		
4. $y = 7(x - 1)^2$	☐ Vertex form ☐ Factored form ☐ Standard form		
5. $y = -x^2 + 12x + 36$	☐ Vertex form ☐ Factored form ☐ Standard form		

6. Rewrite the quadratic equation in factored form.

$$y = x^2 - 3x - 10$$

7. Rewrite the quadratic equation in standard form.

$$y = (x - 3)^2 - 7$$

8. Rewrite the equation in factored form.

$$y = -x^2 + 11x - 28$$

9. Rewrite the equation in standard form.

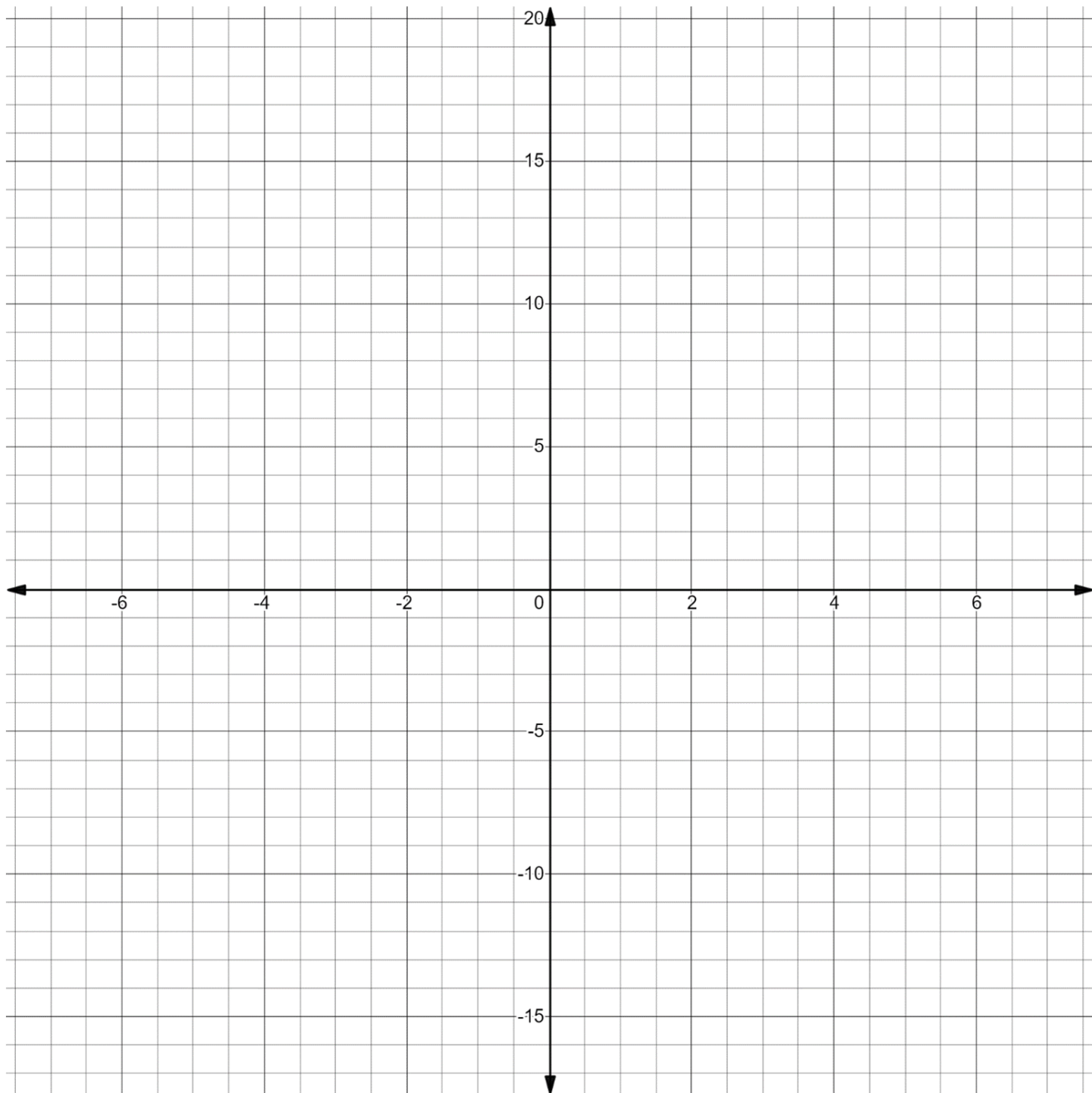
$$y = -2(x - 1)(x + 9)$$

10. Cyrus makes model rockets with his boy scout troop. The rockets launch straight up from 25 meters above the ground with a velocity of 20 meters/second. The height of the rocket is represented by the equation $y = -5x^2 + 20x + 25$, where x is time in seconds. Rewrite the equation in factored form.

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Lesson 4 Practice

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1. Graph the equation $y = 4x^2 - 10x + 8$.



2. Write the key features that can be determined from the equation.

$y = 4x^2 - 10x + 8$	
x-intercept	
y-intercept	
Vertex	
Domain	
Range	
Left end behavior	
Increasing	
Decreasing	
Positive	
Negative	
Opens up/down	
Axis of symmetry	
Right end behavior	

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Re-write each expression into vertex form by completing the square.

1. $x^2 - 10x + 1$

2. $m^2 + 8m - 11$

3. $x^2 - 6x + 9$

4. $j^2 - 2j - 3$

5. $x^2 + 8x - 10$

6. $h^2 + 2h - 3$

7. $q^2 - 12q + 26$

8. $d^2 - 2d - 8$

9. $y^2 + 12x + 20$

10. $u^2 + 16u - 22$

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Lesson 6 Practice

Name _____
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Re-write each expression into vertex form by completing the square.

1. $4t^2 - 16t + 12$

2. $-2x^2 + 8x - 18$

3. $3r^2 - 18r - 12$

4. $-5h^2 - 20h + 3$

5. $3y^2 - 24y - 4$

6. $-2u^2 + 12u + 7$

7. $10n^2 + 40n + 68$

8. $-3d^2 - 18d + 4$

9. $4y^2 + 48y - 15$

10. $9c^2 - 18c - 23$

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Lesson 7 Practice

Name _____
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1. For the expression, the first step to completing the square is shown.

$$x^2 + 18x + 56$$

$$(x^2 + 18x) + 56$$

Finish factoring the perfect square trinomial in vertex form.

In the expression, the value that was added to create the perfect square trinomial is being multiplied by _____, so the value that should be subtracted from 56 to maintain equivalency is _____.

Complete the table to complete the square without using algebra tiles.

	Process	Equation
	Original equation	$y = -5x^2 - 20x + 60$
2.	Factor out the common factor of the first two terms.	$y = \underline{\hspace{1cm}}(x^2 + \underline{\hspace{1cm}}x) + 60$
3.	Determine the value that makes a perfect square trinomial in the parentheses.	$y = \underline{\hspace{1cm}}(x^2 + \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}) + 60$
4.	Then, to maintain equivalence, multiply the factor by the value . Add the opposite of that value .	$y = \underline{\hspace{1cm}}(x^2 + \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}) + 60 + \underline{\hspace{1cm}}$
5.	Rewrite the perfect square trinomial represented in the parentheses as a binomial squared.	$y = \underline{\hspace{1cm}}(x + \underline{\hspace{1cm}})^2 + 60 + \underline{\hspace{1cm}}$
6.	Combine like terms.	$y = \underline{\hspace{1cm}}(x + \underline{\hspace{1cm}})^2 \underline{\hspace{1cm}} \underline{\hspace{1cm}}$

Rewrite each equation in vertex form by completing the square. Then, give the key features revealed by both equations.

7. $y = x^2 - 2x - 15$

8. Key Features

Key Features					
x -intercept		vertex		domain	
x -intercept		increasing		range	
y -intercept		decreasing		opens	
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$			
right end behavior		$x \rightarrow \infty, y \rightarrow \underline{\hspace{2cm}}$			

9. $y = 2x^2 - 8x + 6$

10. Key Features

Key Features					
x -intercept		vertex		domain	
x -intercept		increasing		range	
y -intercept		decreasing		opens	
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The angle, in degrees, at which a batter hits a baseball can affect the distance the ball travels, in feet. The function $B(x) = -0.2x^2 + 14x - 21$ models the distance the baseball travels in ft. depending on the angle, in degrees, at which the ball is hit.

1. What does the variable x represent? Include the units.
2. What does the variable y represent? Include the units.
3. Interpret the y -intercept. Include units.
4. Rewrite the function in vertex form. Show your work.
5. Interpret the vertex in terms of the context. Include units.
6. Complete the table of values for $B(x)$.

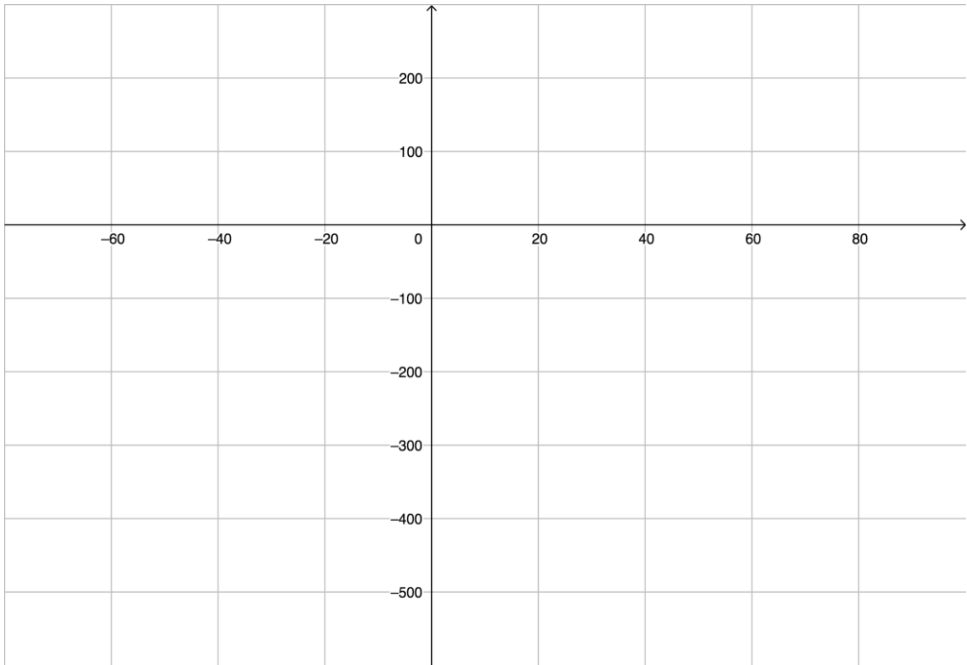
x	-2	0	2	7	17	23	35	68
$B(x)$								

7. Circle the values in the table that do not make sense in terms of the context.

8. Complete as many of the key features as possible using the given function, the function rewritten in vertex form, and the table of values. Do not use approximate or rounded values.

Key Features					
x -intercept		vertex		domain	
x -intercept		increasing		range	
y -intercept		decreasing		opens	
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$			
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9. Graph the function.



10. Use the graph, table of values, or function to determine the reasonable constraints for the context. Write the constraints in set builder notation.

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Florida Real Estate Team just bought a 100 unit apartment building and plans to rent all of the apartments out. The function $P(u) = -10u^2 + 1760u - 50,000$ represents the profit earned, P , depending on the amount of apartment units, u , they rent.

1. Rewrite the function in vertex form.

Complete the statements.

2. The

- ☐ vertex
 - ☐ factored
 - ☐ standard

 form of the function could be used to determine the interval on which the real estate company would experience a profit.

3. The

- ☐ vertex
 - ☐ factored
 - ☐ standard

 form of the function can be used to determine when the real estate company experiences the greatest profit.

4. The company is predicted to experience the greatest profit when _____ units are rented, resulting in a profit of \$_____.

Adam is coding a new game where darts are launched at a moving target. He enters the function $H(t) = -0.5t^2 + 6t - 10$ to model the height, H , of the darts in relationship to the ground after t seconds.

5. Rewrite the function in vertex form.

6. Rewrite the function in factored form.

Complete the statements.

7. The

☐ vertex
☐ factored
☐ standard

 form of the function can be used to determine how long the dart is above the ground. The dart is in the air for _____.

8. The dart was launched from a point _____ units

☐ above
☐ below

 the ground and went

☐ above
☐ below

 ground after _____ seconds.

9. The

☐ vertex
☐ factored
☐ standard

 form of the function can be used to determine when the dart reaches its highest point.

10. The dart reaches its highest point of _____ units _____ seconds after it is launched.

Algebra 1
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Lesson 9 Practice

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☐ above
☐ below

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